

WARDROBE+

A Virtual Closet and Outfit Planner Mobile Application Using Flutter And Firebase

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Course – Bachelor's Degree in Information Technology

Institution: Mount Kenya University

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INTRODUCTION

Wardrobe+ is a virtual closet and outfit planner mobile application developed using Flutter and Firebase. The application is designed to help users digitally organize their clothing items, create outfit combinations, and manage their wardrobe efficiently through a modern mobile platform.

With the increasing use of smartphones and digital lifestyle applications, many individuals seek easier ways to organize their clothing and improve outfit planning. Traditional wardrobe management methods are often inconvenient and time-consuming, leading to repetitive outfit choices and poor clothing organization.

WARDROBE+ provides users with a smart and user-friendly solution where they can upload clothing items, categorize outfits, save favorite combinations, and access their wardrobe digitally at any time. The application also demonstrates the use of modern mobile development technologies such as Flutter for frontend development and Firebase for backend services, authentication, and cloud storage.

The project aims to provide practical experience in mobile application development, user interface design, Firebase integration, and software project documentation while creating a system with real-world commercial potential.

PROBLEM STATEMENT

Many individuals face challenges organizing their clothing items and planning outfits efficiently. Traditional wardrobe management methods make it difficult for users to keep track of available clothing combinations, resulting in repetitive dressing patterns, poor outfit coordination, and wasted time when selecting clothes.

Additionally, there are limited mobile applications that provide simple and user-friendly digital wardrobe management systems targeted at helping users organize clothing items and plan outfits effectively.

Therefore, there is a need for a smart mobile application that enables users to digitally manage their wardrobe, create outfit combinations, and improve clothing organization through an accessible and modern platform.

OBJECTIVES

General objective

To develop a virtual closet and outfit planner mobile application that enables users to digitally organize clothing items and manage outfit combinations efficiently.

Specific objective

1. To design and develop a user-friendly mobile application using Flutter.
2. To implement user authentication using Firebase Authentication.
3. To allow users to upload and categorize clothing items digitally.
4. To enable users to create and save outfit combinations.
5. To integrate Firebase for cloud data storage and management.
6. To design a responsive and modern user interface for improved user experience.
7. To test and evaluate the functionality and usability of the application.

REQUIRMENTS ANALYSIS

Functional Requirements

The system should allow users to:

- register and log into the application
- upload clothing items and images
- categorize clothing items
- create and save outfit combinations
- edit or delete wardrobe items
- store user data using Firebase

Non-Functional Requirements

The system should:

- provide a user-friendly interface
- operate efficiently on Android devices
- securely authenticate users
- maintain reliable cloud data storage
- provide fast response to user interactions

Software Requirements

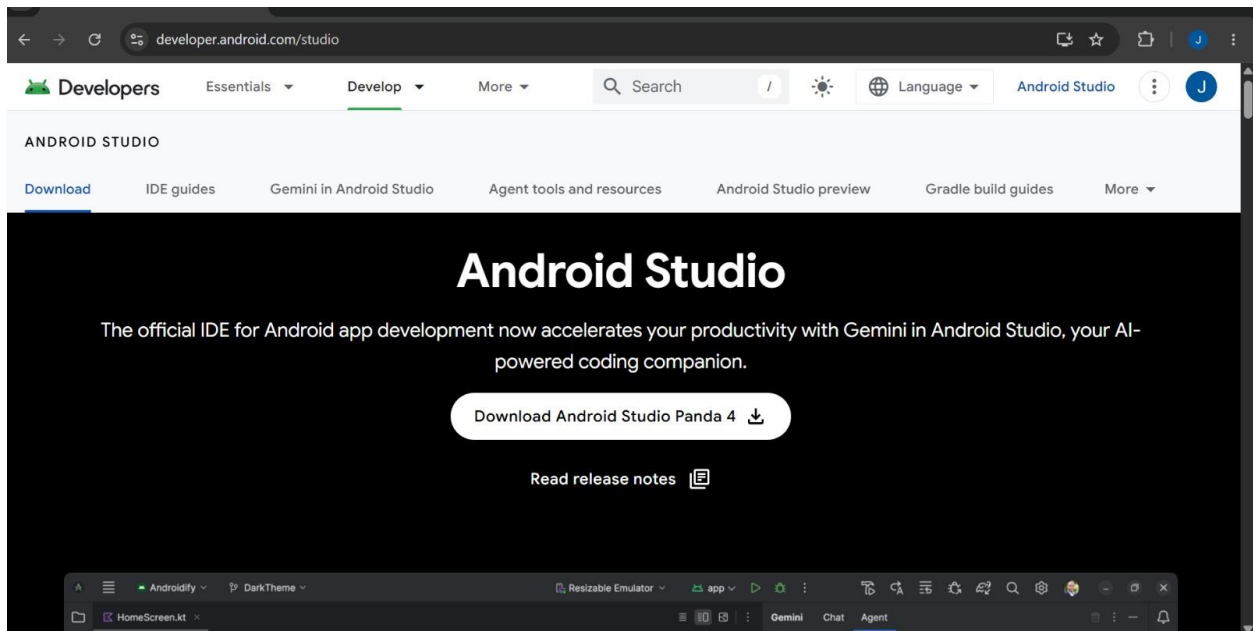
Software	Purpose
Flutter	Mobile app development
Android Studio	Development environment
Firebase	Authentication, database, storage
Figma	UI/UX design
GitHub	Version control

INSTALLATION AND SETUP PROCESS

Android Studio

1. Android Studio Installation

Android Studio was installed as the main development environment together with Android SDK, Android Emulator, Flutter Plugin and Dart Plugin





Android Studio

Welcome to Android Studio Setup

Setup will guide you through the installation of Android Studio.

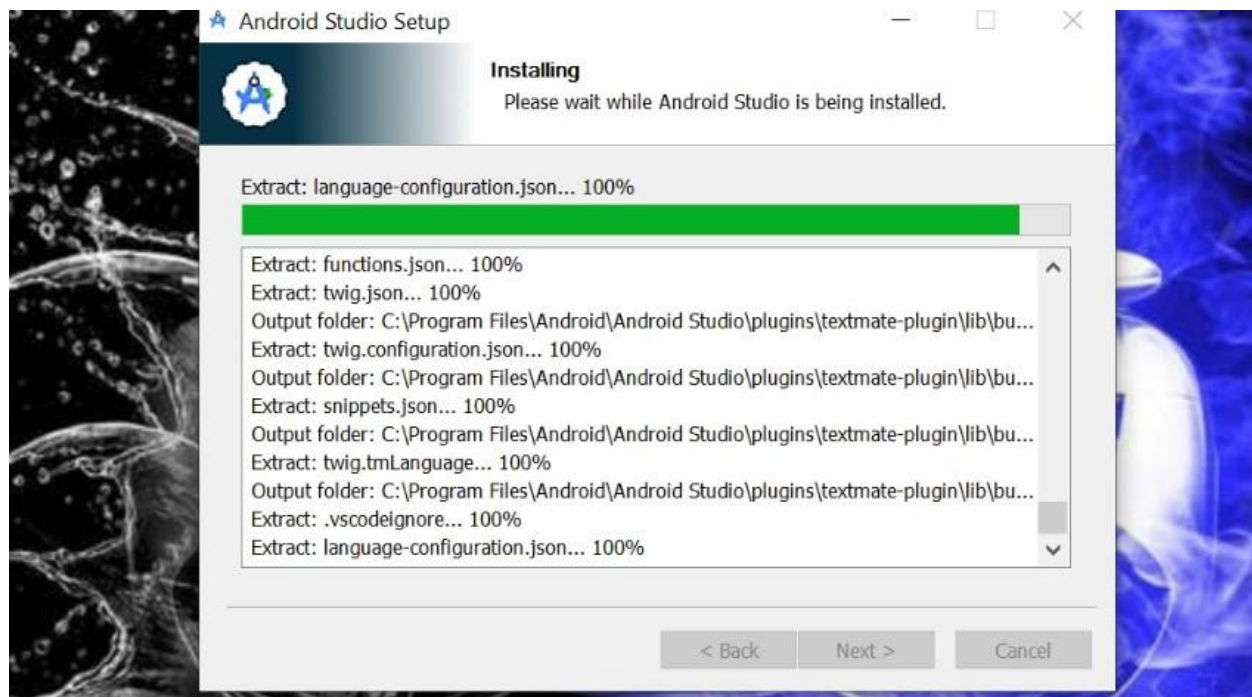
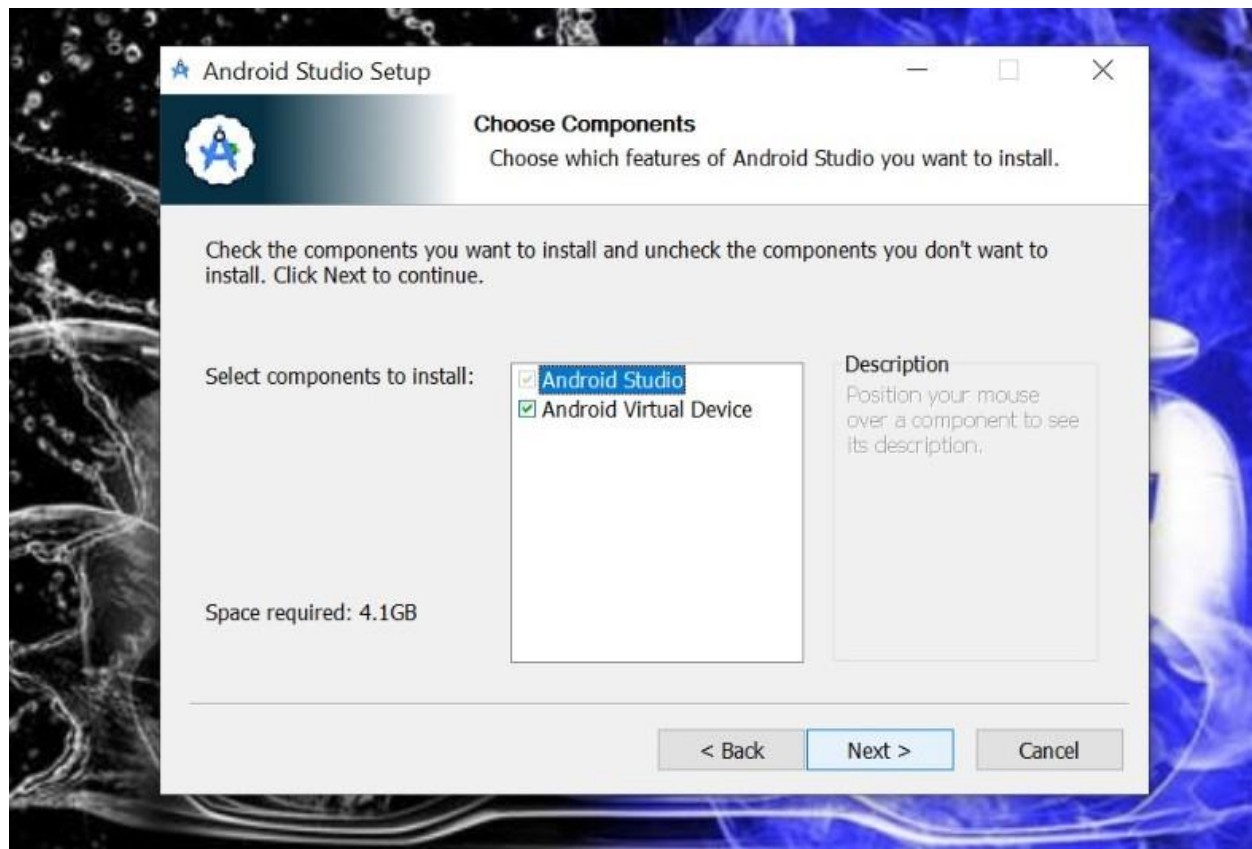
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.

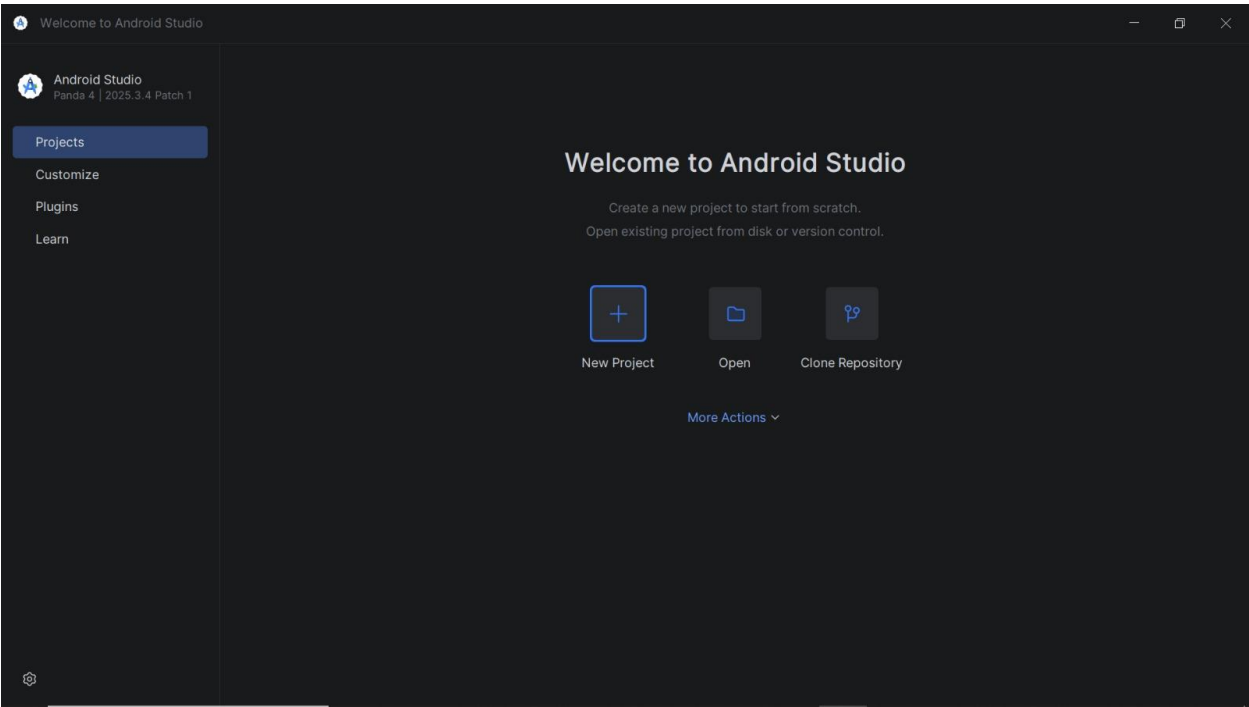
Click Next to continue.

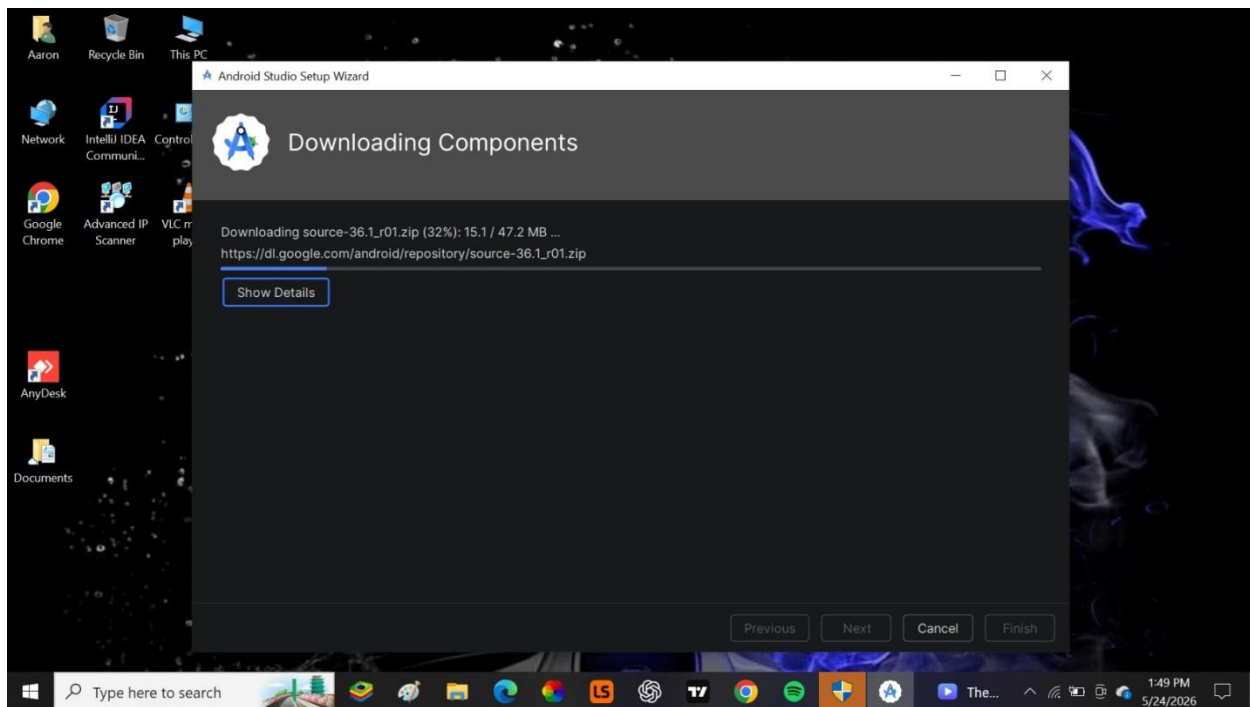
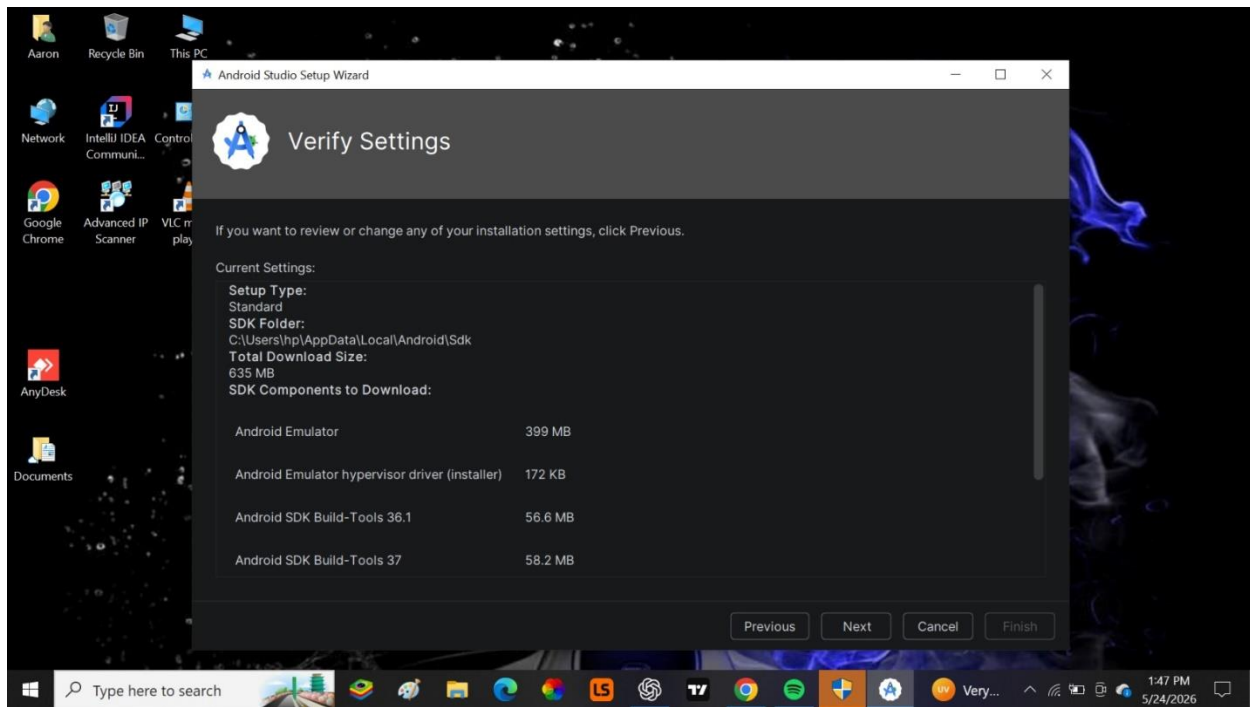
< Back

Next >

Cancel







Flutter plugin

Programming Language

Flutter

[Google](#) [Plugin homepage](#) ↗

Restart IDE 92.0.0

Overview

What's New

Reviews

Additional Info

Support for developing Flutter applications. Flutter gives developers an easy and productive way to build and deploy cross-platform, high-performance mobile apps for both Android and iOS. Installing this plugin will also install the Dart plugin.

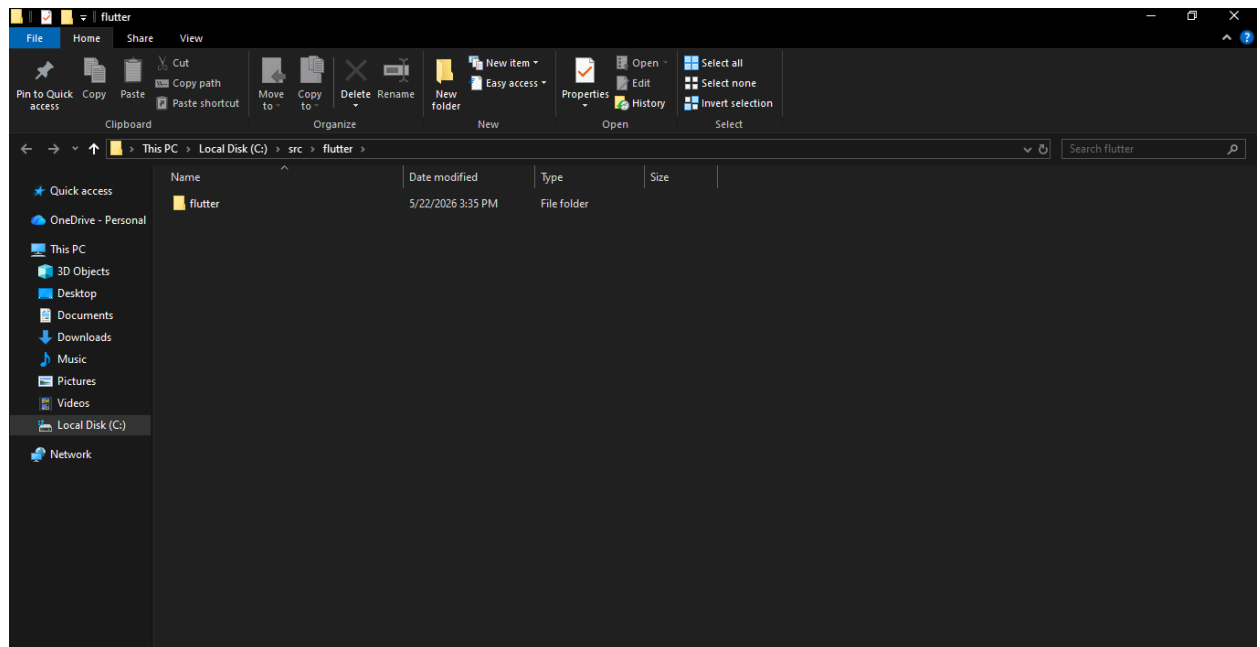
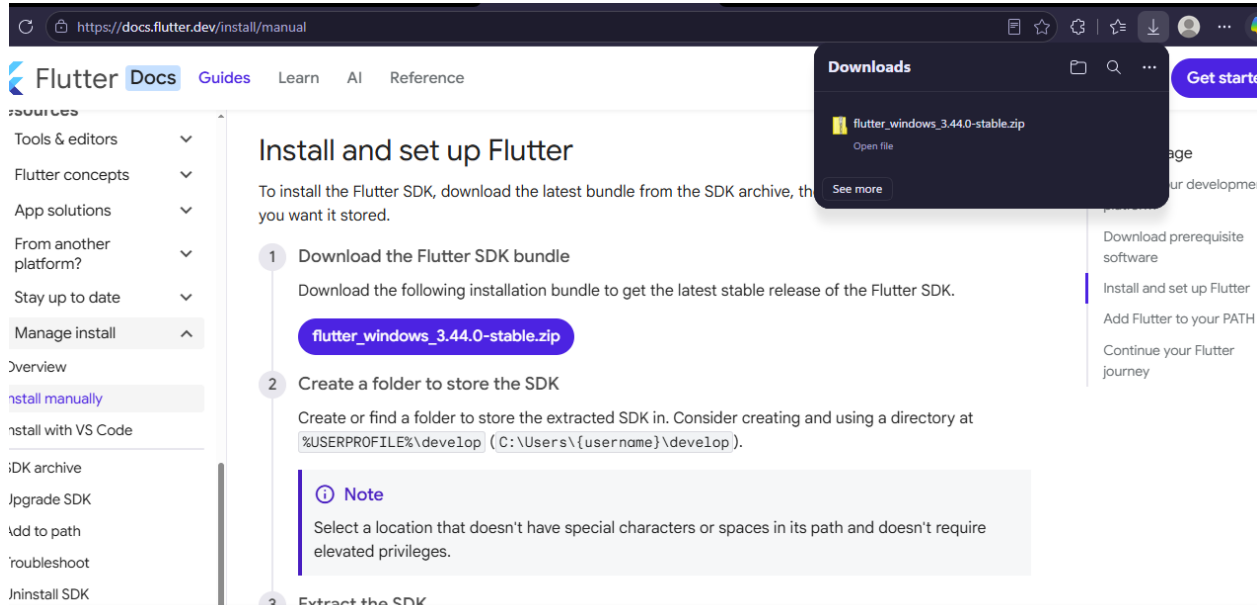
For some tools, this plugin uses Chromium through JxBrowser to display content from the web. JxBrowser complies with LGPL and offers an option to replace Chromium with another component. To do this:

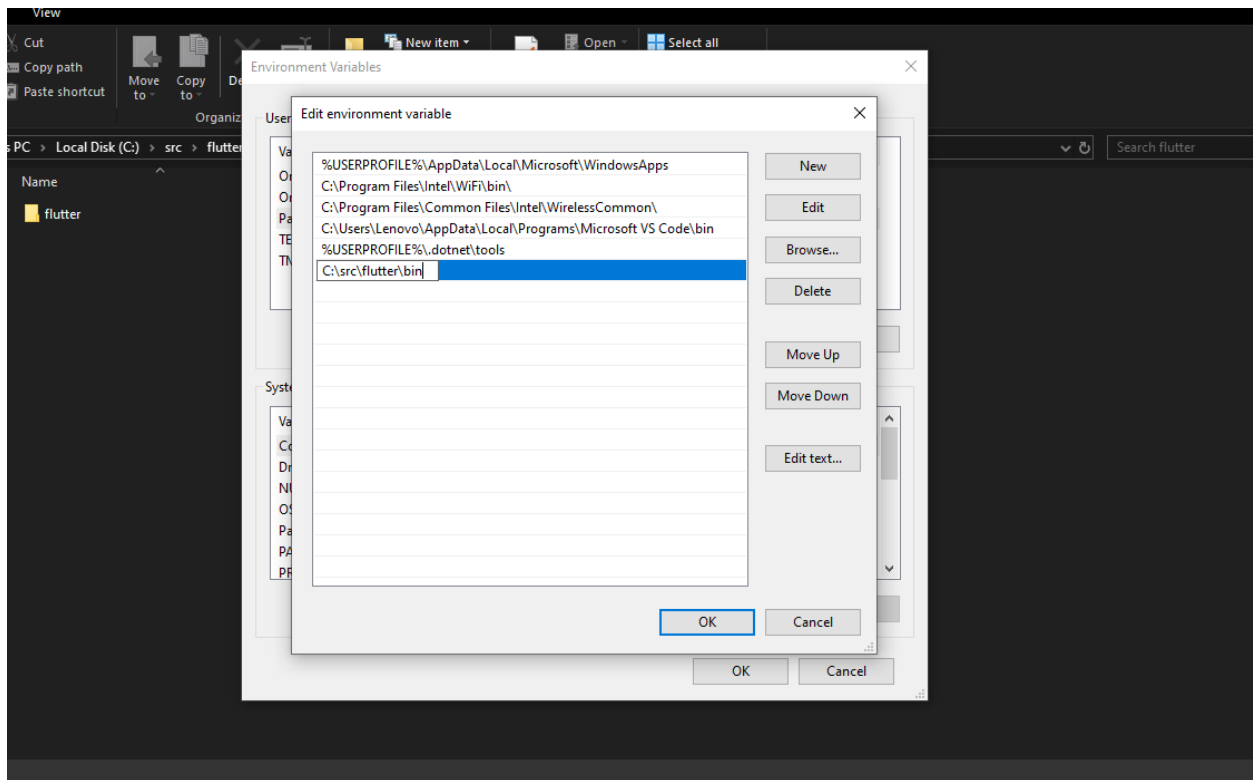
- Find the JxBrowser files stored in the [plugins directory](#), under `/flutter-intellij/jxbrowser`.
- The LGPL requirements are at [from JxBrowser](#), here you can also download the build script to relink with modified components.

Flutter setup

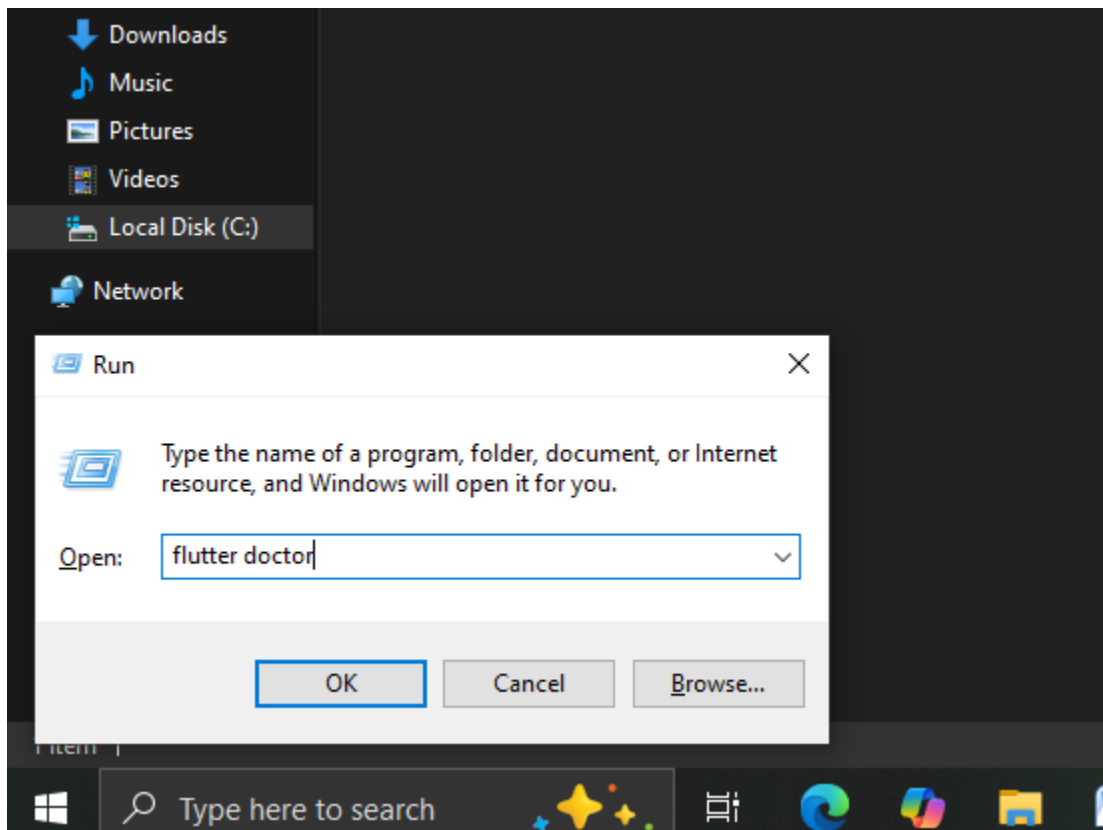
Flutter SDK was downloaded and installed to enable cross-platform mobile application development.

Flutter environment was configured by adding Flutter to the system PATH variable.





Installation was verified by running flutter doctor;



```
Microsoft Windows [Version 10.0.19045.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>flutter doctor

Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.44.0, on Microsoft Windows [Version 10.0.19045.5737], locale en-US)
[✓] Windows Version (10 Pro 64-bit, 22H2, 2009)
[!] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
    ✗ cmdline-tools component is missing.
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio#command-line-tools-only and make sure
      to set the ANDROID_HOME environment variable.
      See https://developer.android.com/studio/command-line for more details.
    ✗ Android license status unknown.
      Run `flutter doctor --android-licenses` to accept the SDK licenses.
      See https://flutter.dev/to/windows-android-setup for more details.
[✓] Chrome - develop for the web
[!] Visual Studio - develop Windows apps (Visual Studio Community 2022 17.14.32)
    ✗ Visual Studio is missing necessary components. Please re-run the Visual Studio installer for the "Desktop
      development with C++" workload, and include these components:
        MSVC v142 - VS 2019 C++ x64/x86 build tools
        - If there are multiple build tool versions available, install the latest
        C++ CMake tools for Windows
        Windows 10 SDK
[✓] Connected device (3 available)
[✓] Network resources

! Doctor found issues in 2 categories.

C:\Users\Lenovo>
```

Appearance & Behavior

Appearance

Menus and Toolbars

System Settings

File Colors

Scopes

Notifications

Data Editor and Viewer

Quick Lists

Required Plugins

Trusted Locations

Path Variables

Presentation Assistant

Keymap

Editor

Plugins

Version Control

Build, Execution, Deployment

Languages & Frameworks

C/C++

Android SDK

JVM Logging

Kotlin

Languages & Frameworks > Android SDK Revert changes

Manager for the Android SDK and Tools used by the IDE

Android SDK Location: C:\Users\Lenovo\AppData\Local\Android\Sdk Edit Optimize disk space

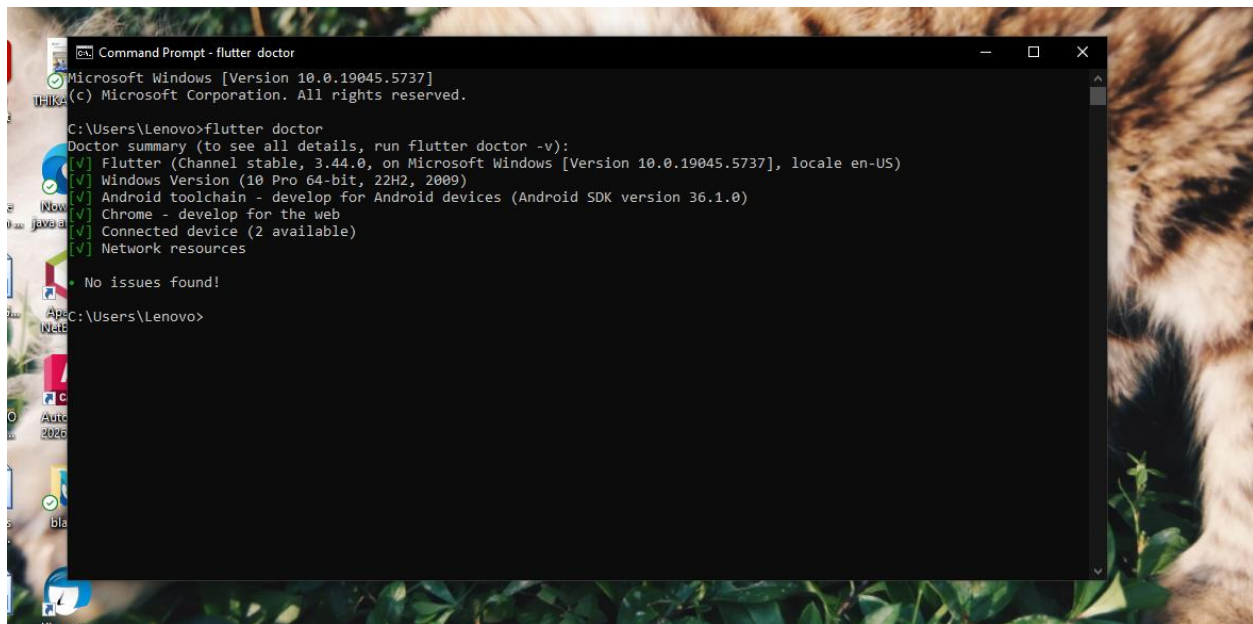
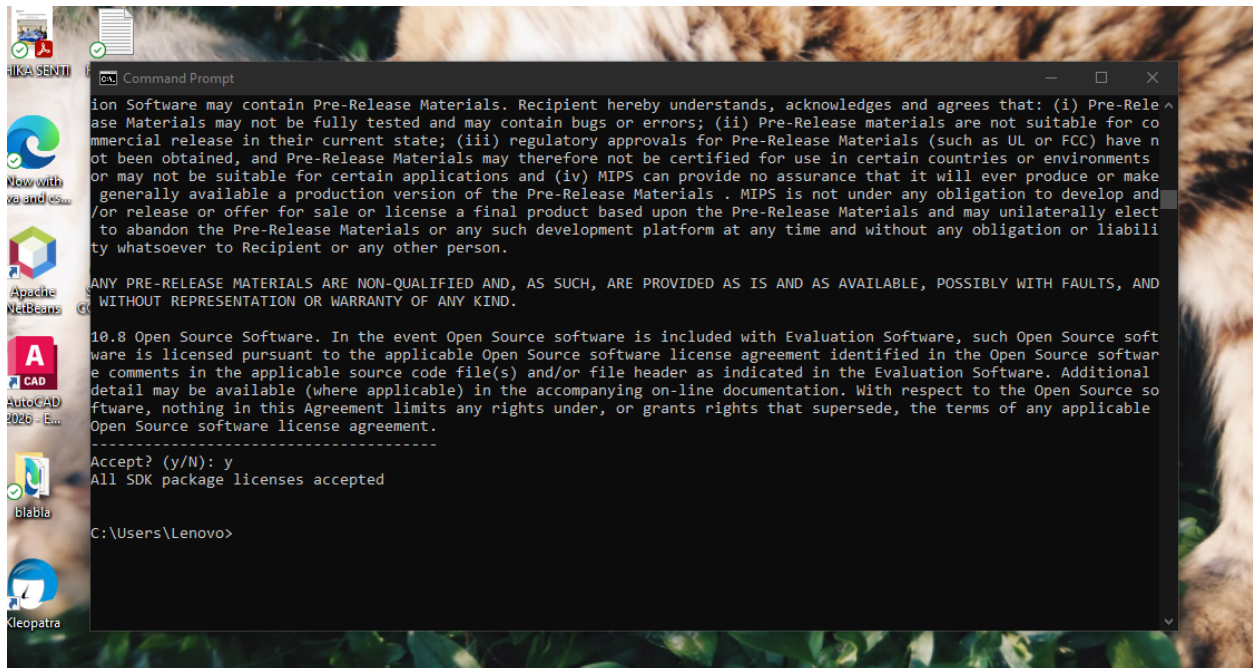
SDK Platforms SDK Tools SDK Update Sites

Below are the available SDK developer tools. Once installed, the IDE will automatically check for updates. Check "show package details" to display available versions of an SDK Tool.

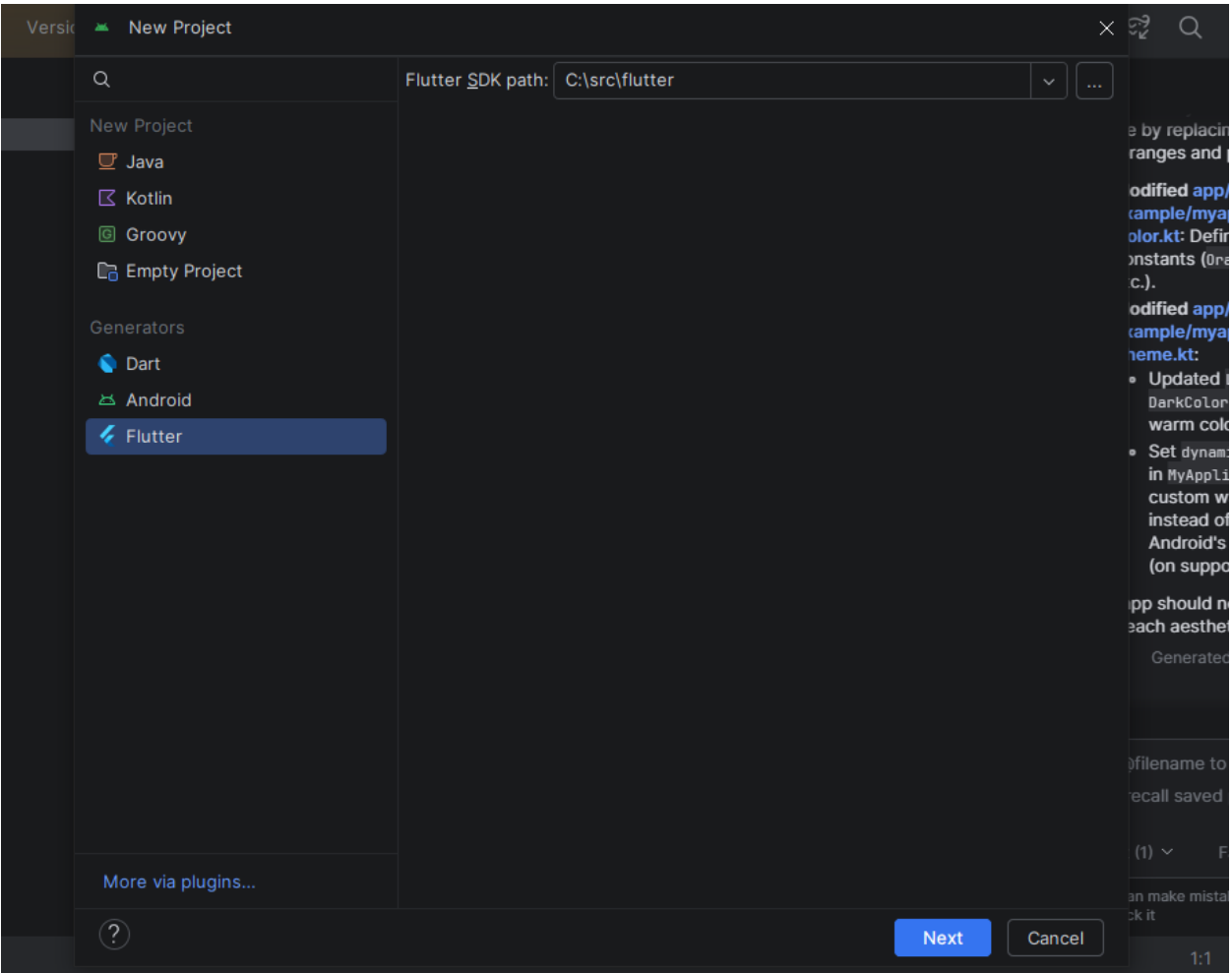
Name	Version	Status
☒ Android SDK Build-Tools		Installed
☐ NDK (Side by side)		Not Installed
☒ Android SDK Command-line Tools (latest)		Installed
☐ CMake		Not Installed
☐ Android Auto API Simulators	1	Not installed
☐ Android Auto Desktop Head Unit Emulator	2.0	Not installed
☒ Android Emulator	36.5.11	Installed
☒ Android Emulator hypervisor driver (installer)	2.2.0	Installed
☐ Android Project Templates	0.1.0	Not installed
☒ Android SDK Platform-Tools	37.0.0	Installed
☐ Android Support Repository	47.0.0	Not installed
☐ Google Play APK Expansion library	1	Not installed
☐ Google Play Instant Development SDK (Deprecated)	1.9.0	Not installed
☐ Google Play Licensing Library	1	Not installed

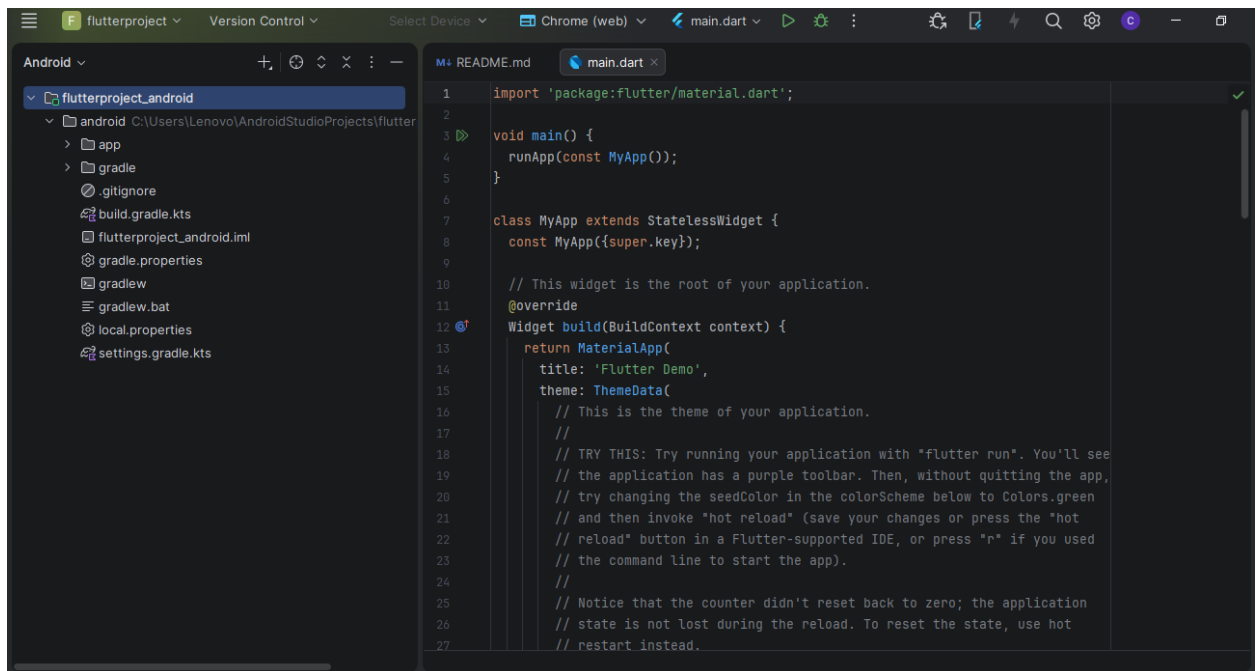
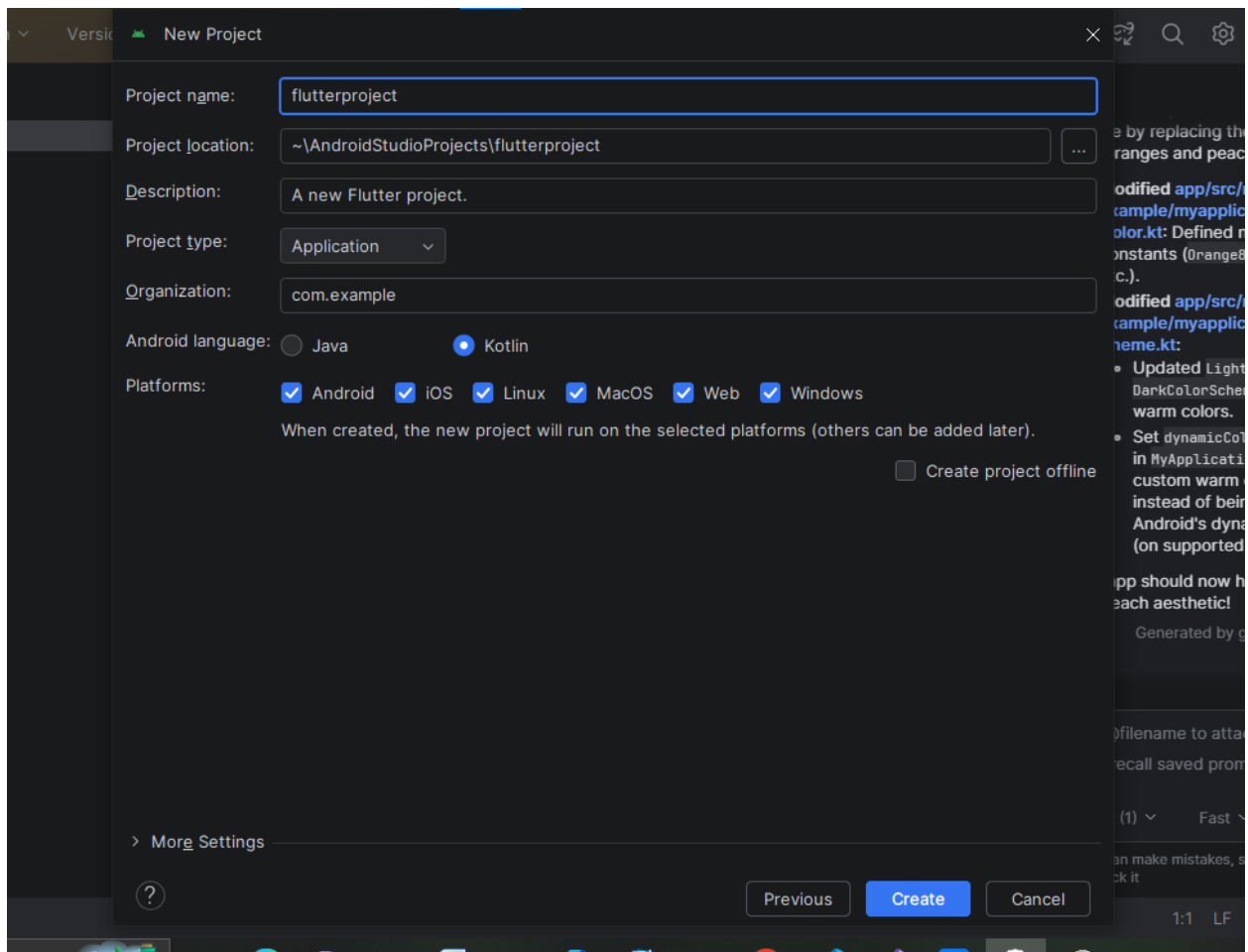
☒ Hide Obsolete Packages ☐ Show Package Details by editing and lines of code... not show again

OK Cancel Apply



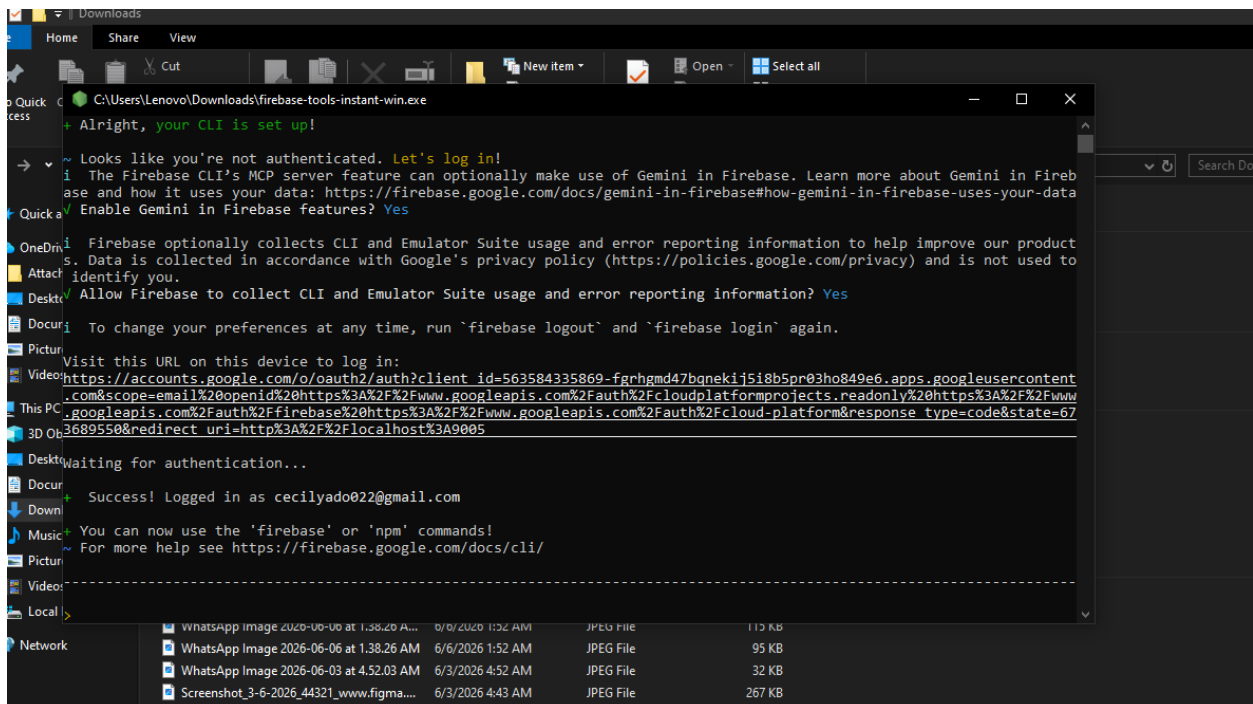
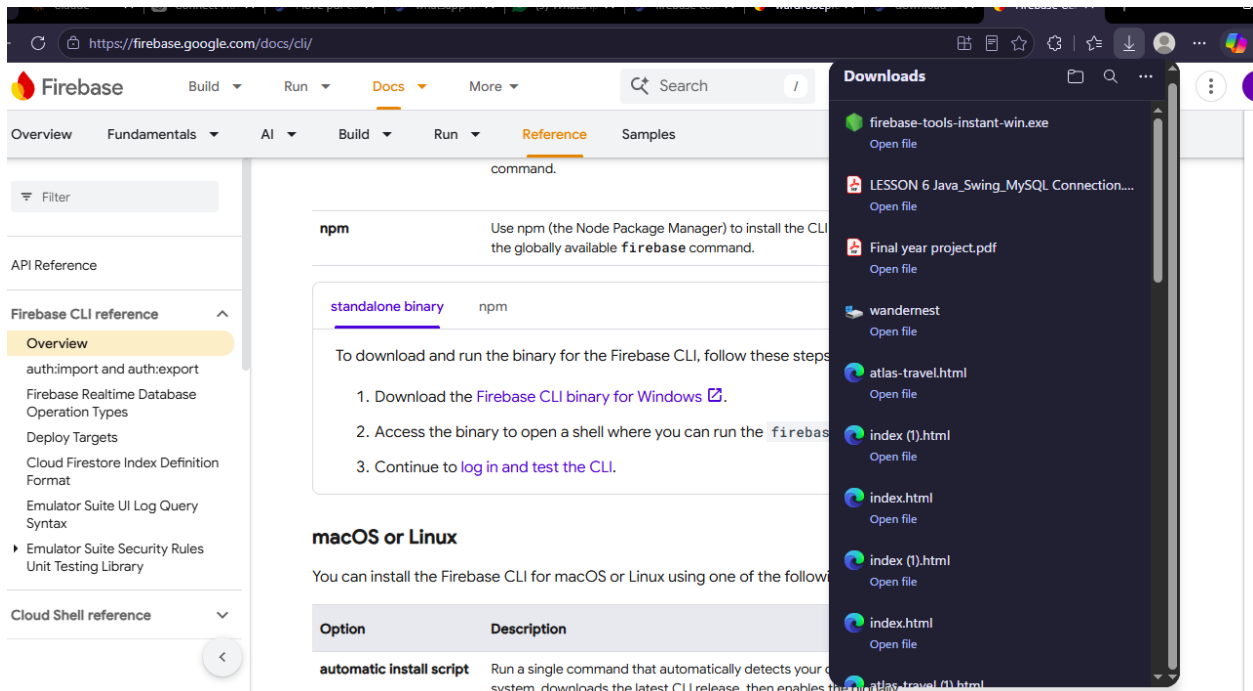
Creation of flutter project

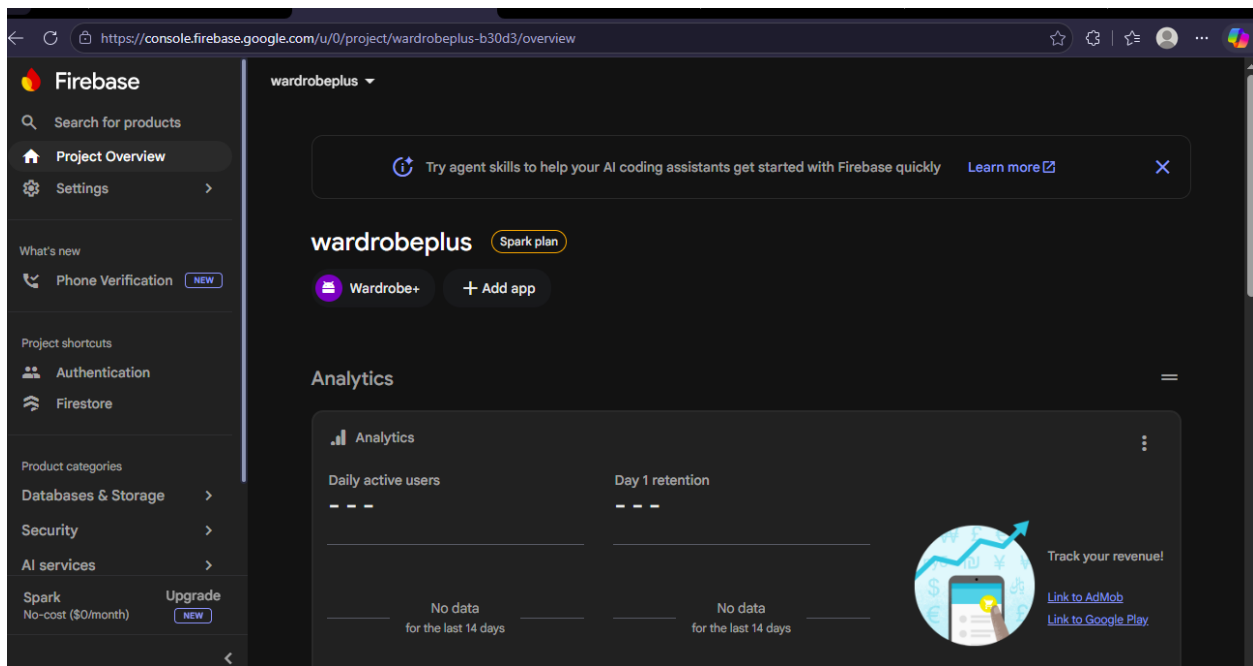




Firestore Setup

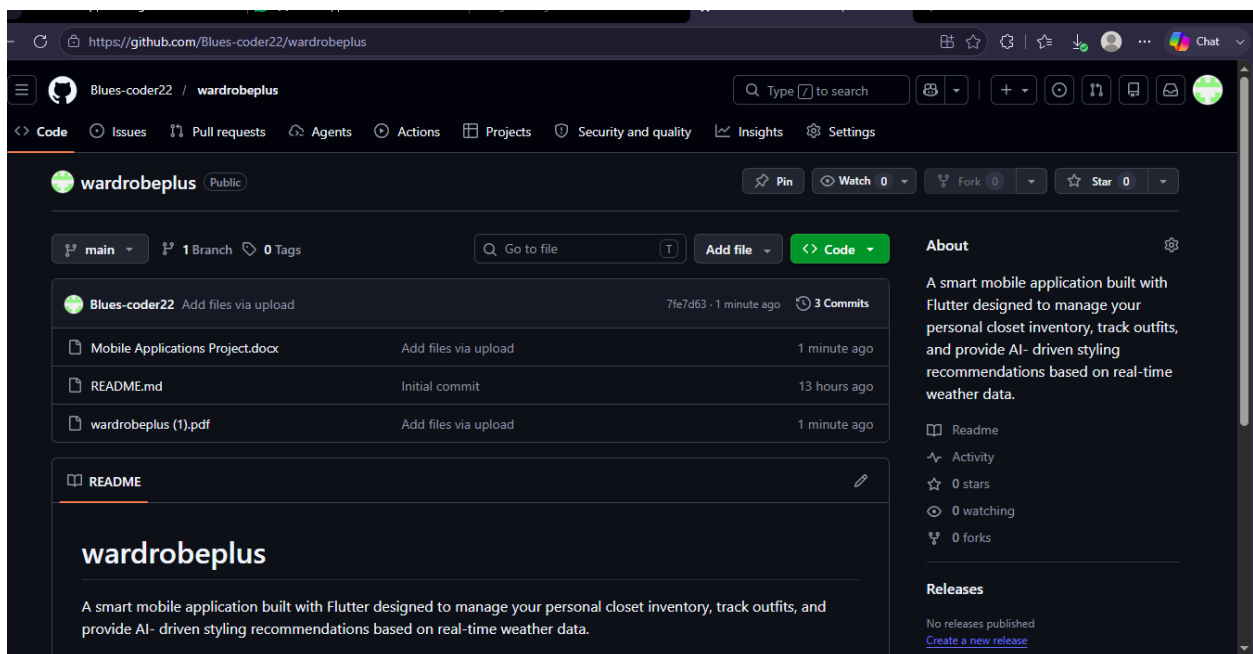
Firestore was used for authentication and database (Firestore).





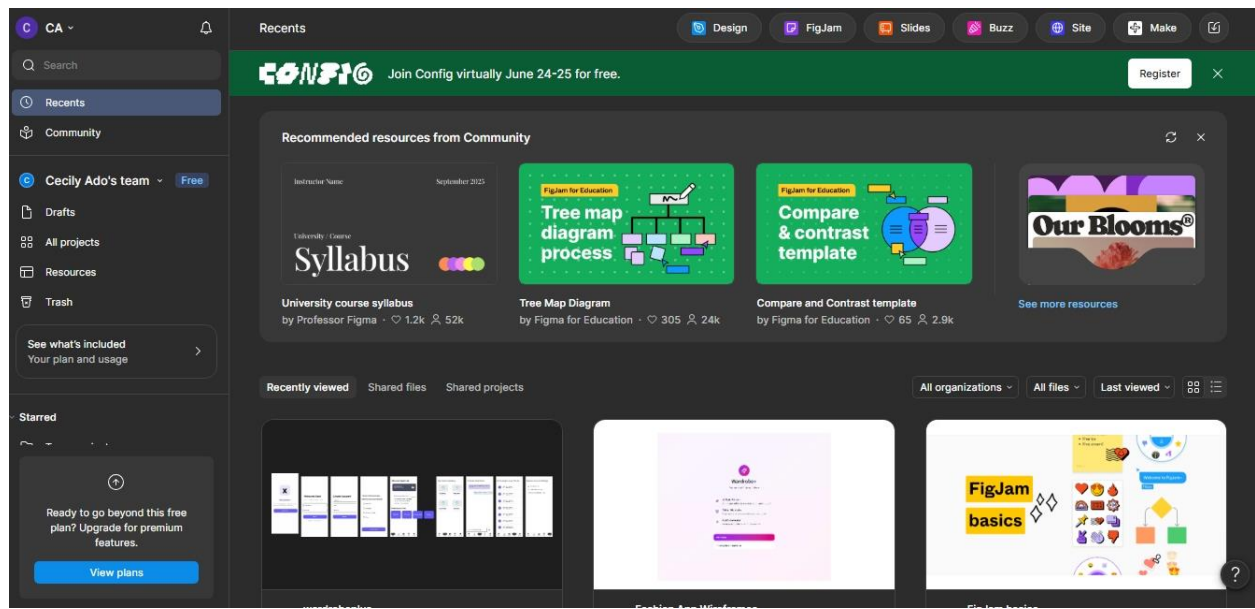
Github setup

GitHub was used for version control and project backup.



Figma Setup

Figma was used to create wireframes and UI designs for the application.



Emulator Testing

An Android emulator was configured for testing the application during development using:
flutter run. Used a phone for wireless debugging.

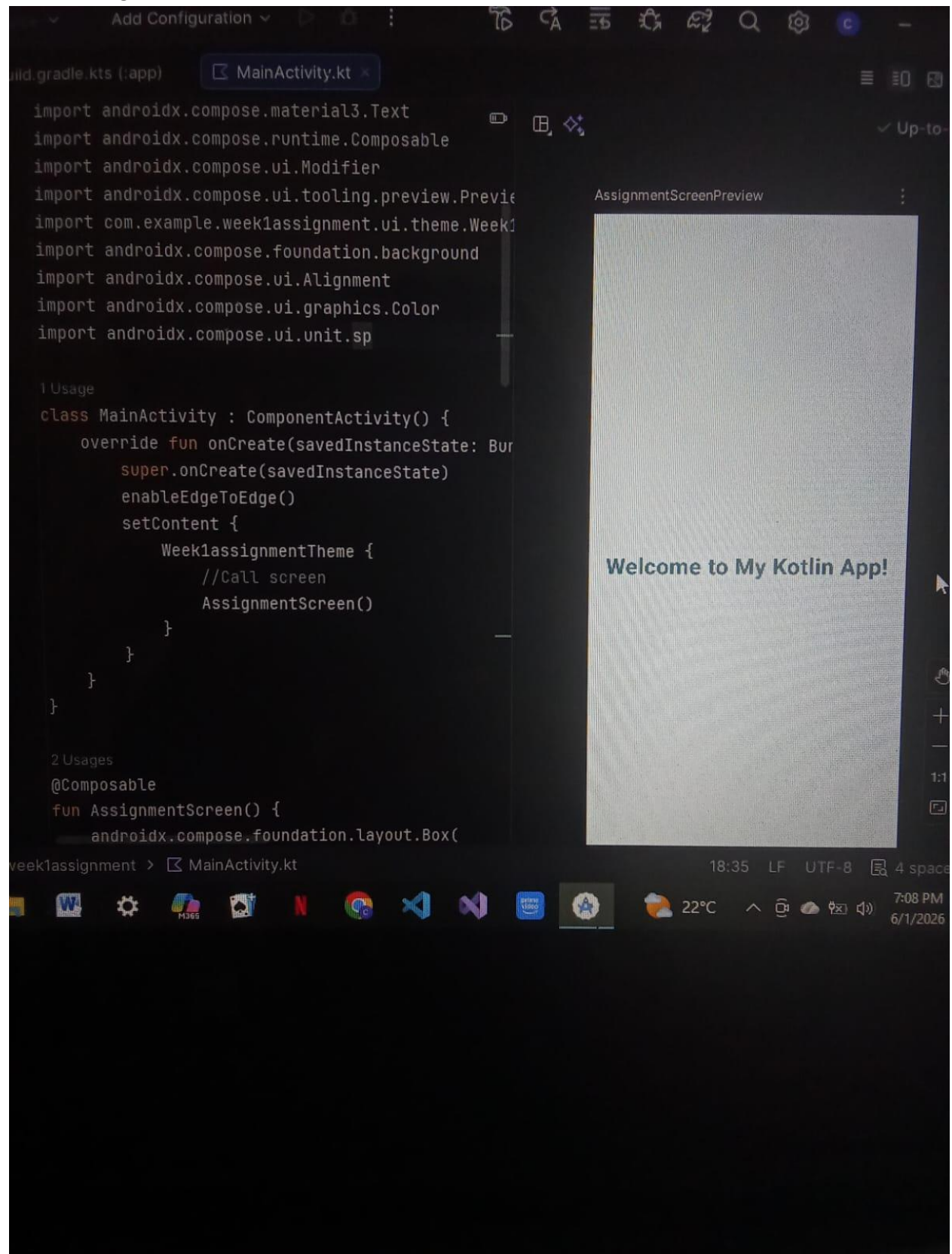


Wardrobe+

Your AI-powered digital closet. Organize layouts, track items, and style smart outfits cleanly.

Get Started

Week 1 Assignment



The code is in kotlin.

Mainactivity.kt file

```
package com.example.weeklassignment

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.foundation.layout.padding
import androidx.compose.material3.Scaffold
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.tooling.preview.Preview
import com.example.weeklassignment.ui.theme.WeeklassignmentTheme
import androidx.compose.foundation.background
import androidx.compose.ui.Alignment
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.unit.sp

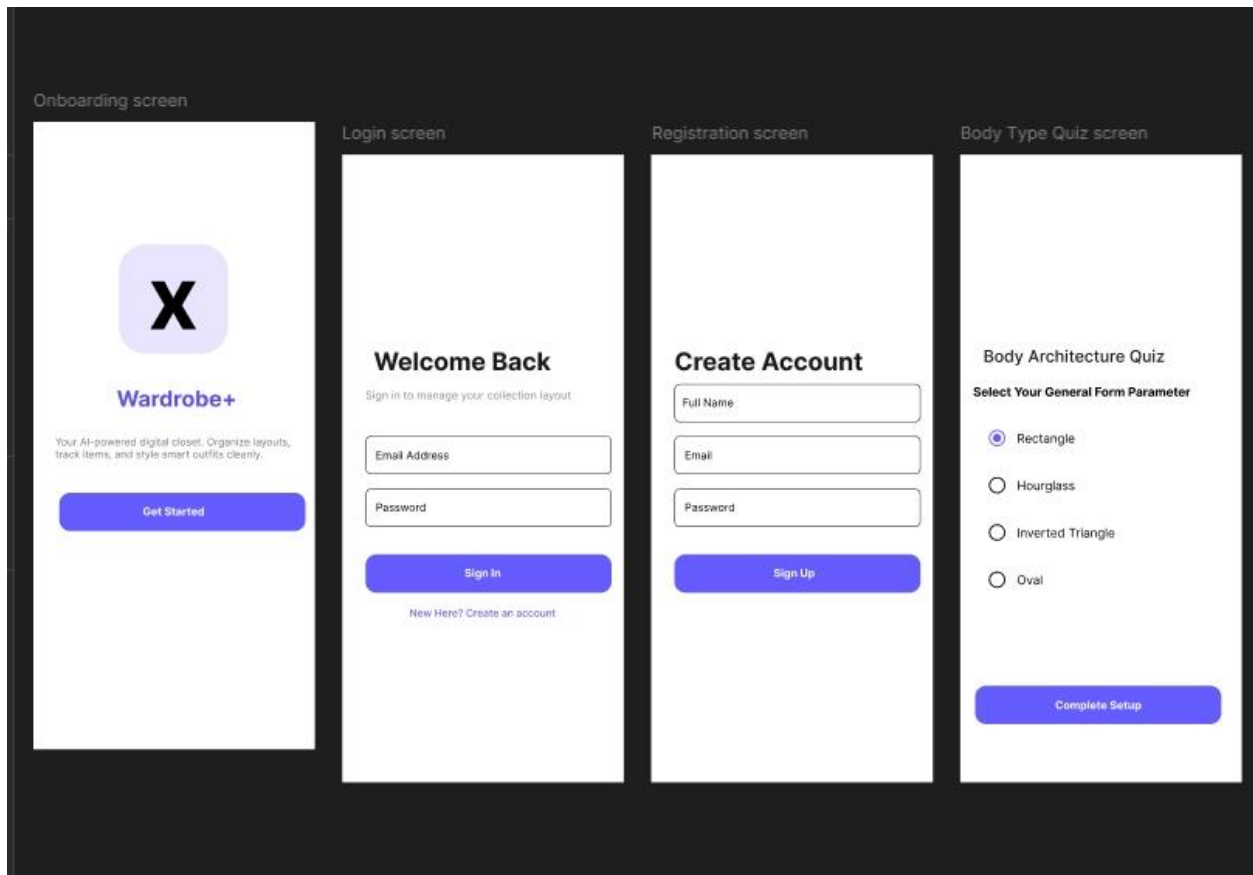
class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            WeeklassignmentTheme {
                //Call screen
                AssignmentScreen()
            }
        }
    }
}

@Composable
fun AssignmentScreen() {
    androidx.compose.foundation.layout.Box(
        modifier = androidx.compose.ui.Modifier
            .fillMaxSize()
            .background(androidx.compose.ui.graphics.Color(0xFFE0F7FA)), //
        Background Color
        contentAlignment = androidx.compose.ui.Alignment.Center
    ) {
        Text(
            text = "Welcome to My Kotlin App!", // App Title
            fontSize = 28.sp, // Font Size
            fontWeight = androidx.compose.ui.text.font.FontWeight.Bold,
            color = androidx.compose.ui.graphics.Color(0xFF006064)
        )
    }
}

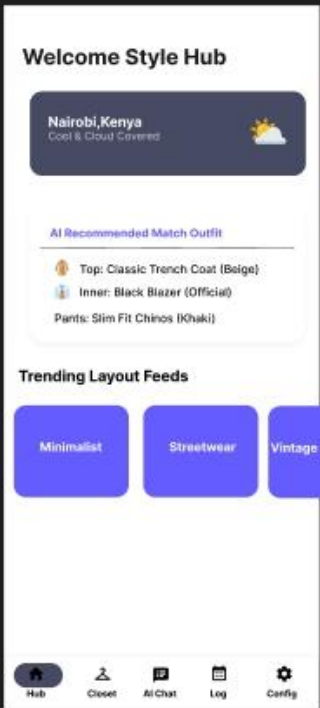
@Preview(showBackground = true)
@Composable
```

```
fun AssignmentScreenPreview() {
    WeeklassignmentTheme {
        AssignmentScreen()
    }
}
```

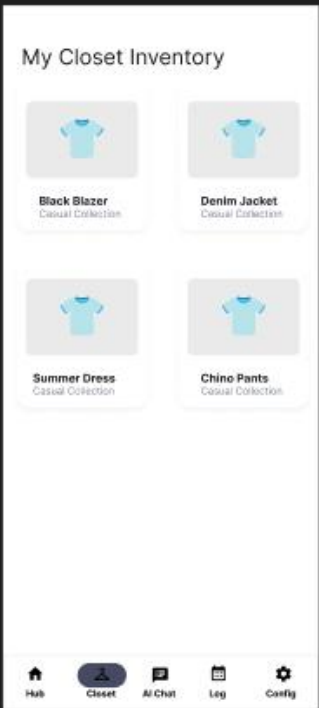
UI-WIREFRAMES



Dashboard



Wardrobe screen



chat screen



Chronological Log Planner

1 Day 1 Outfits Block
Worn: Black Blazer Setup

2 Day 2 Outfits Block
Worn: Black Blazer Setup

3 Day 3 Outfits Block
Worn: Black Blazer Setup

4 Day 4 Outfits Block
Worn: Black Blazer Setup

5 Day 5 Outfits Block
Worn: Black Blazer Setup

6 Day 6 Outfits Block
Worn: Black Blazer Setup

7 Day 7 Outfits Block
Worn: Black Blazer Setup

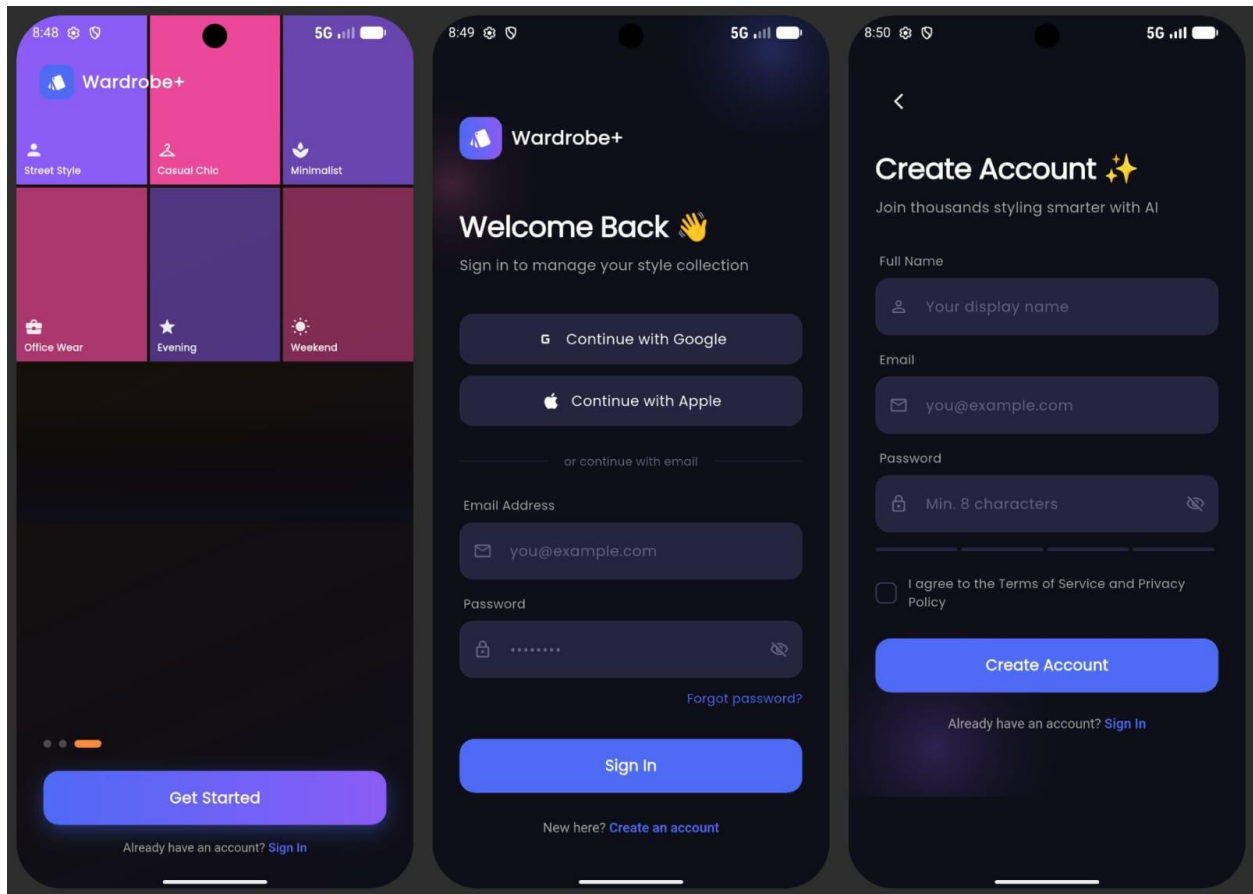
System Account Settings

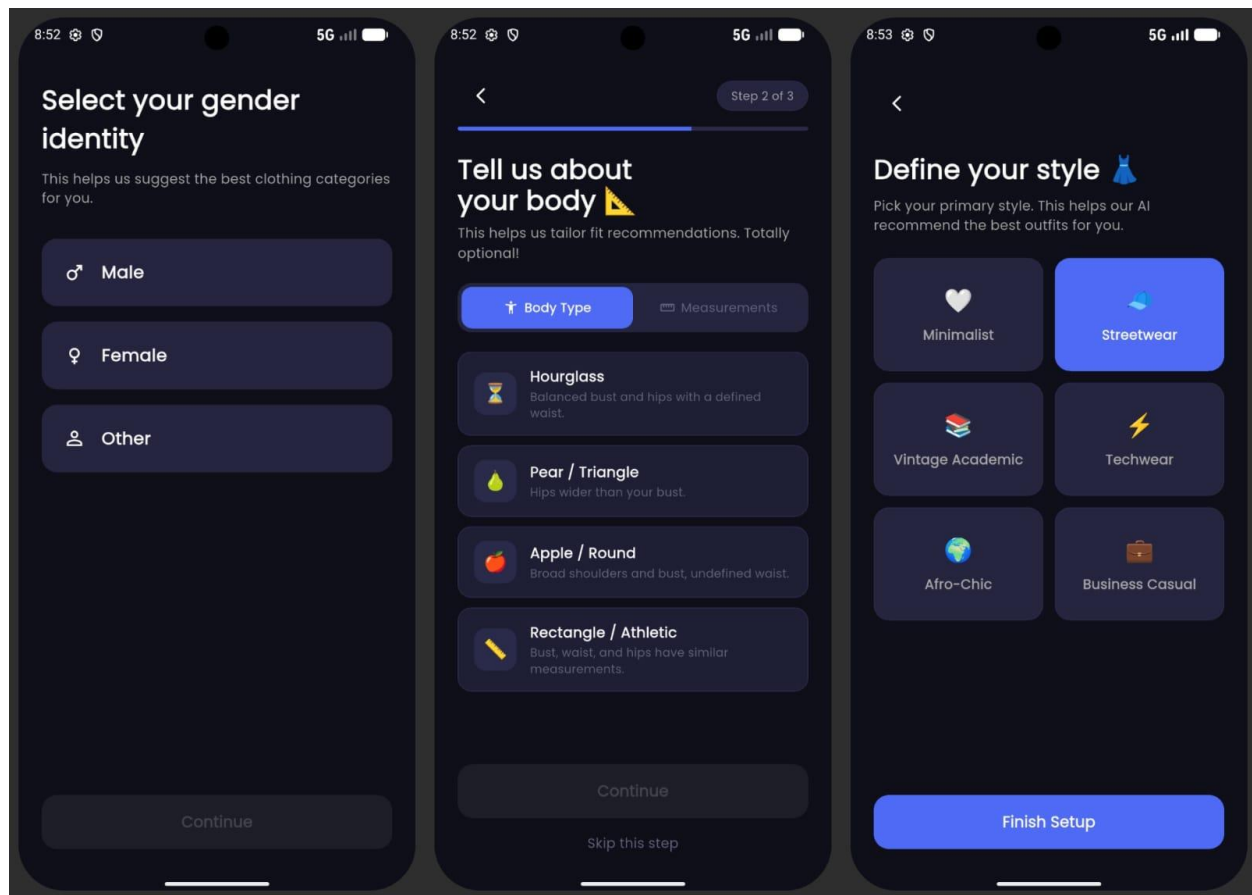
User Profile Node

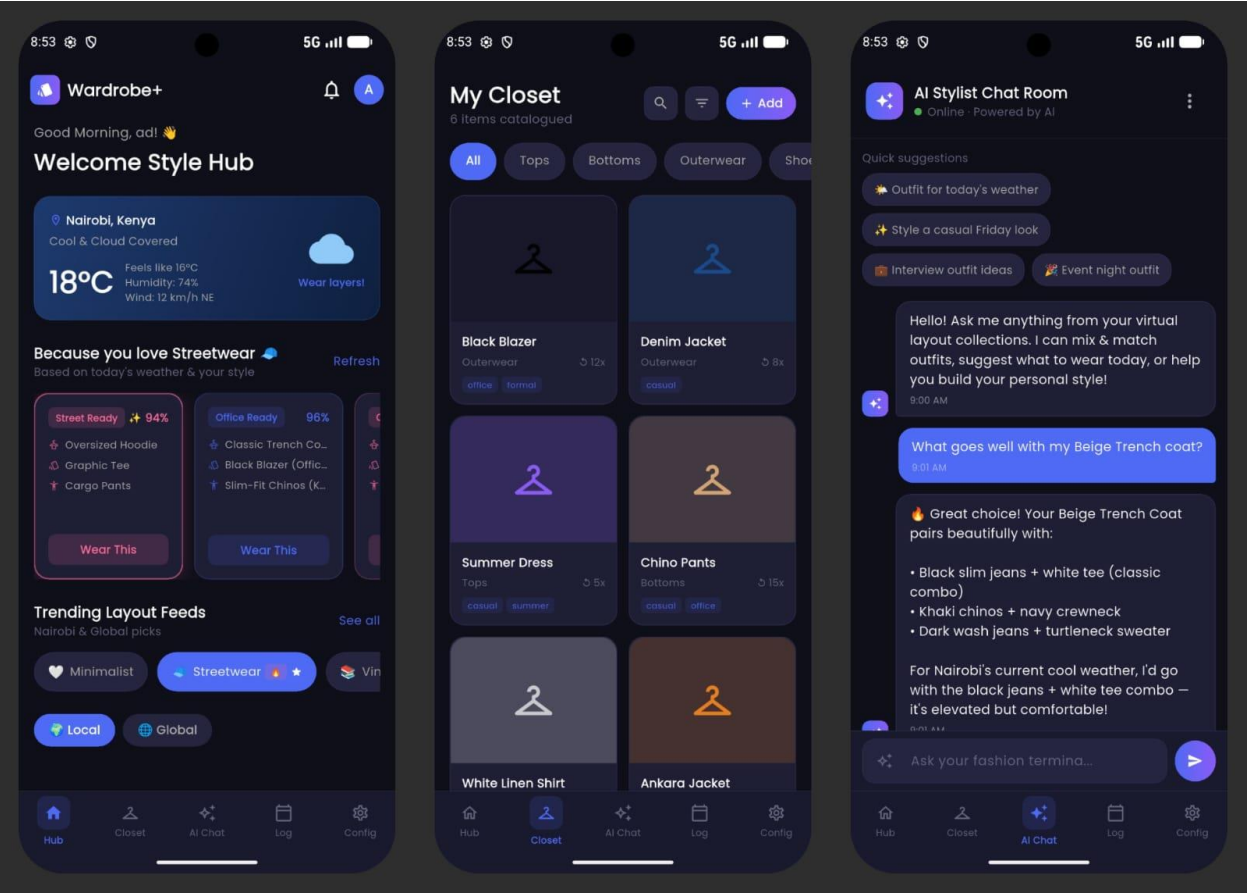
Modify Style Chips Layout

Force Dark Palette Engine ☒

GUI DESIGN







Chronological Log Planner

June 2026

5 Logged

2 Planned

71% Unique

1

Day 1 Outfits Block

Black Blazer + White Shirt
Mon, Jun 2



2

Day 2 Outfits Block

Denim Jacket + Jeans
Tue, Jun 3



3

Day 3 Outfits Block

Ankara Jacket + Chinos
Wed, Jun 4



4

Day 4 Outfits Block

Trench Coat + Slim Fit
Thu, Jun 5



5

Day 5 Outfits Block

Linen Shirt + Chinos
Fri, Jun 6



6

Day 6 Outfits Block

Casual Tee + Jeans
Sat, Jun 7



+ Log Outfit

7



Hub



Closet



AI Chat



Calendar



Confia

System Account Settings



AM

Amara Mwangi

amara@email.com

Edit

Afro-Chic



24

Items



38

Outfits



52

Days Logged

User Profile Node

Edit Profile



Body Type Profile

Hourglass



Style Preferences



Location Settings

Nairobi, KE



Modify Style Chips Layout

Custom Style Tags



AI Style Sensitivity

High



Trend Region Focus

Multiple Global



Hub



Closet



AI Chat



Log



Settings